



UNIVERSITY OF LAGOS

Chemical Engineering Department

2025/2026 Session, 2nd Semester

CHG 569 - Environmental Pollution Engineering



*"Technology is a queer thing, it brings you great gifts with one hand,
and stabs you in the back with the other"*

- C. P. Snow

Introduction

As mankind enjoys the benefits of technology/industrialization, it is very important to examine the other side of the coin - the adverse effects of technology/industrialization. CHG 569 is a course that looks at this other side of the coin, and discusses measures to be taken to avoid, and where this is not possible, to reduce the impact of 'the stab in the back'. It is about making the environment a better place to live in.

Our Common Future, also known as the **Brundtland Report** (in recognition of former Norwegian Prime Minister Gro Harlem Brundtland's role as Chair of the World Commission on Environment and Development (WCED)), was published in 1987 by the United Nations through the Oxford University Press, and the term Sustainable development was defined as development that meets the needs of the present without compromising the ability of future generations to meet their own needs. – [overview of report attached](#)

In 1997, 18 international institutions representing the chemical engineering profession globally signed the London Communiqué ([attached](#)) which pledged '*to make the world a better place for future generations*'

In 2001, 20 chemical engineering institutions signed the Melbourne Communiqué ([attached](#)) committing each of them to work towards a shared global vision based on sustainable development at the World Congress on Chemical Engineering.

25 years on the environment is still an issue

📅 Workplan for 2025/2026 Session

University Calendar

Commencement of Lectures:	Monday, April 27	
End of Lectures:	Friday, July 31.	(14 wks)
Lecture free week	Monday, August 3 – Friday, August 8	(1 wk)
2nd Semester Examinations:	Monday, August 10 – Saturday, August 29	(3 wks)
End of Session:	Saturday, September 4	

📅 Overall Aim

To develop students' understanding of the environmental problem and provide them with the fundamental scientific and engineering principles for control of pollutants in the environment, with emphasis on designing systems and strategies that ensure long-term environmental sustainability.

📅 Course Objectives

At the end of this course students should:

1. Appreciate the fact that there is an environmental problem due to advancement in technology and industrialization, occasioned by human population growth.
2. Be able to design effective environmental management strategies for air pollution, industrial wastewater pollution and solid waste nuisance.
3. Propose and adopt strategies towards preventing/reducing environmental harm and ensuring environmental sustainability
4. Hold their heads high, in confidence of the fact that they can contribute their quota to making the world a better place for all to live in, thus ensuring sustainable development.

Course Calendar

Basis: 13 weeks of lectures

Week	Course Outline	
2 07/05/2026	Module I - Introduction to the Environmental Problem Pollution and the Environment Fiction and Facts (A fable & True Stories: Bhopal Tragedy, Erin Brockovich & Tragedy at Love Canal)	
3 14/05/2026	Module II - Air Pollutants and their effects (Local and Global)	Test I
4 21/05/2026	Module III - Energy and the Environment Module IV - Legislation	Test II
5 28/05/2026	Module V - Methods of Air pollution Control	
6 04/06/2026		
7 11/06/2026		
8 18/06/2026		
9 25/06/2026	Module VI - Industrial Wastewater Management Sources and Characterization Wastewater Management options	
10 02/07/2026		Test IV
11 09/07/2026	Module VII - Solid Waste Management	Test V
12 16/07/2026	Module VIII - Engineering for sustainable Development	
13 23/07/2026	- Biomimicry for sustainable innovations	Test VI
14 30/07/2026	Module IX - Other Topics in Environmental Pollution Control - Environmental Ethics, - Noise Pollution, - Radiation Pollution - Digital Pollution	
		Test VII

 Course slides

Slides for the modules above can be downloaded from [Get Slides](#)

Course Resources (Optional)

- Introduction to Environmental Engineering by Davis and Cornwell, McGraw-Hill
- Environmental Engineering by Howard S. Peavy, Donald R. Rowe and George Tchobanoglous, McGraw-Hill
- Environmental Pollution and Control in Chemical Process Industries by S.C. Bhatia, Khanna Publishers, Delhi
- Industrial Pollution Prevention Handbook by Harry M. Freeman, McGraw-Hill
- The Hitch Hiker's Guide to LCA by Baumann & Tillman, Studentlitteratur
- Sustainable Industrial Design and Waste Management: Cradle-to-cradle for Sustainable Development by Salah El-Haggar
- **And most importantly the lecture slides for the course.**

👉 Course Time-table

Day/Time of the week

Thursday: 9:00 – 11:00

👉 Course Venue (for physical lectures)

Chemical Engineering Classroom 01

👉 Course Zoom link (for Virtual lectures)

Click [Zoom Link](#)

Meeting ID: 833 9484 9123

Passcode: CHG569

👉 Course Grading Scheme

	Bonus
Class Participation	
Tutorial	10
Test(s)	30
Term Paper	10
Final Exam	50
Total	<u>100</u>

👉 Course Lecturer

Tolu Ajayi

tajayi@unilag.edu.ng

0803-301-0847

Room 513 4th floor, old building

Attachments

Our Common Future: Brundtland Report OVERVIEW

In 1983, the United Nations Secretary-General invited Norwegian prime minister Gro Harlem Brundtland to chair a World Commission on Environment and Development. Concern about the acute pressure of population growth, modern technology and consumer demand on the planetary fabric had been smoldering away since the 1970s. Now a new generation of environmental worries - global warming, deforestation, species loss, toxic wastes - had begun to capture scientific and popular attention. The world's natural resources were being rapidly depleted, often in the name of development, but the poverty this development was supposed to correct was as widespread as ever.

By the time the Brundtland Commission delivered its report on Our Common Future in 1987, population growth was no longer seen as the major threat to the harmony of the planet. Almost all of it was among poorer people. And it was not they who were consuming the Earth's supply of fossil fuels, warming the globe with their carbon emissions, depleting its ozone layer with their CFCs, poisoning soil and water with their chemicals, or wreaking ecological havoc with their oil spills. In fact, their consumption of the world's resources was minute compared to that of the industrialized world.

Brundtland declared that poverty in the developing world was less cause than effect of contemporary environmental degradation, the outcome of insensitive technology transfer that pauperized people and natural systems. If all the world's people were to live like North Americans, a planet four times as large would be needed. Only 'sustainable' development could blend the fulfillment of human needs with the protection of air, soil, water and all forms of life - from which, ultimately, planetary stability was inseparable.

Thus the concept of 'sustainable development' was launched: social and economic advance to assure human beings a healthy and productive life, but one that did not compromise the ability of future generations to meet their own needs. Sustainable development brought environmentalism into poverty reduction and poverty reduction into environmentalism in a single and simple formula. It led to the first Earth Summit - the UN Conference on Environment and Development - at Rio de Janeiro in 1992, and to the formulation of Agenda 21.

Chapter 1: A Threatened Future

Chapter 2: Towards Sustainable Development

Chapter 3: The Role of the International Economy

Chapter 4: Population and Human Resources

Chapter 5: Food Security: Sustaining the Potential

Chapter 6: Species and Ecosystems: Resources for Development

Chapter 7: Energy: Choices for Environment and Development

Chapter 8: Industry: Producing More with Less

Chapter 9: The Urban Challenge

Chapter 10: Managing the Commons

Chapter 11: Peace, Security, Development, and the Environment

Chapter 12: Towards Common Action: Proposals for Institutional and Legal Change

THE LONDON COMMUNIQUÉ

We, the representatives of 18 societies representing chemical engineers world-wide and acting here in our personal capacities, subscribe to the following statement:

THE KEY CHALLENGE FOR OUR PROFESSION IN THE TWENTY-FIRST CENTURY IS:

To use our skills to improve the quality of life: foster employment, advance economic and social development, and protect the environment.

This challenge encompasses the essence of sustainable development.

We will work to make the world a better place for future generations.

SPECIFICALLY, CHEMICAL ENGINEERS WILL:

Design processes and products which are innovative, energy-efficient and cost-effective, make the best use of scarce resources and ensure that waste and adverse environmental impact are minimised.

Achieve the highest standards of safety in making and using products of all kinds.

Provide the processes and products which give the people of the world shelter, clothing, food and drink, and which keep them in good health.

Work with other disciplines to seek solutions.

Engage in honest and open dialogue with the public on the challenges presented by manufacture of the products which the public requires.

Promote research to allow the profession to respond fully to global demands.

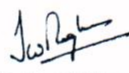
Encourage the brightest and best young people into the profession, and promote lifelong professional development.

Therefore we must co-operate together and recognise each other's efforts in striving to meet this challenge.

We acknowledge that this challenge cannot be met by our efforts alone, but this does not lessen the responsibility to pursue it.

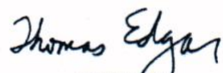
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10 APRIL 1997

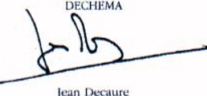

Kondapuram Raghavan
Indian Institute of Chemical Engineers


Gerhard Kreysa
DECHEMA


Jim Snow
Institution of Engineers, Australia

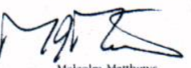

Thomas Edgar
American Institute of Chemical Engineers


Gurli Mogensen
Society of Danish Engineers


Jean Decaure
Groupe Français de Génie des Procédés


Nikolai Kulov
Mendeleev Russian Chemical Society


Tomas Masek
Czech Society of Chemical Engineering


Malcolm Matthews
Hong Kong Institution of Engineers

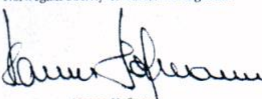

Chris Albertyn
South African Institution of Chemical Engineers

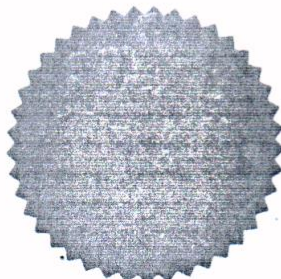

Shintaro Batusaki
Society of Chemical Engineers, Japan


Gyula Körövelyessy
Hungarian Chemical Society


Alistair Miller
Canadian Society for Chemical Engineering


Knut Henriksen
Norwegian Society of Chartered Engineers


Hanns Hofmann
European Federation of Chemical Engineering




Vol Bamelis
Gesellschaft für Verfahrenstechnik und Chemieingenieurwesen, VDI


Simon Prins
Nederlandse Ingenieursvereniging


Keith Taylor
Institution of Chemical Engineers

THE MELBOURNE COMMUNIQUÉ

We, the representatives of twenty organisations representing chemical engineers world-wide, subscribe to the following statement:

Entering the Twenty-First Century, we in the chemical engineering profession renew our commitment to using our skills to strive to improve the quality of life, foster employment, advance economic and social development and protect the environment through sustainable development.

Chemical engineering uses the principles of science to develop and provide technologies that improve the lives of people everywhere. From chemical engineering come materials: minerals, metals, ceramics, polymers, paper and composites. Food production and processing and water purification rely on chemical engineering. To enhance the health of the world's peoples, chemical engineering makes possible vaccines, pharmaceuticals and biotechnology. Shelter, clothing, transport and information technology all rely on chemical engineering. We develop and deliver the energy resources on which communities depend.

In meeting society's needs we are committed to designing processes and products that are innovative, energy-efficient and cost-effective, and that make the best use of scarce resources. We are committed to the highest standards of personal and product safety. We seek to eliminate waste and adverse environmental effects in the development, manufacture, use and eventual disposal of the products of society.

In an increasingly global society, we are committed to meeting the collective needs of the world's growing population while working to avoid and eliminate practices that are unsustainable. We recognise the profound concern relating to climate change. We must promote a better understanding of the complex science surrounding climate change, while also striving to develop and implement sound technologies to mitigate its effects.

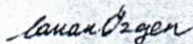
We commit to developing public understanding of the challenges and choices facing our world, and the role chemical engineering can play. We will use our talents, knowledge and organisational skills for the continued betterment of humanity to protect the public welfare. We will practice our profession according to its high ethical standards. We will promote lifelong professional development and will enthuse the brightest young people so that they enter our profession.

We will take this Communiqué to our members, reinforcing their awareness of their obligations to serve society. We also commit to working with industries, governments, universities and other organisations that shape the future of the world.

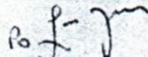
We acknowledge both our professional responsibilities and the need to work with others as we strive to meet the challenges facing the world in the Twenty-First Century.

AGREED IN MELBOURNE AT THE SIXTH WORLD CONGRESS OF CHEMICAL ENGINEERING

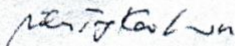
27th September, 2001



Canan Öngen
Chemical Engineering Society of Turkey



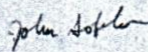
Po Lock Yue
Hong Kong Institute of Engineers



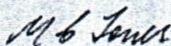
Prier Ting Kai Wu
Chinese Institute of Chemical Engineers
(Chinese Taipei)



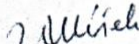
U. W. Felch
DIXCHEMA Society of Chemical
Engineering & Biotechnology



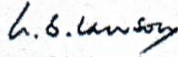
John Sofrenko
American Institute of Chemical Engineers



Mark Tozer
Institution of Chemical Engineers in Australia



Tomáš Mělek
Czech Society of Chemical Engineering



Graham Lawson
Institution of Chemical Engineers



Harald Müller
South African Institution of Chemical Engineers



Russell Burton
Asian Pacific Confederation of Chemical Engineering



Roberto Andrade
Inter-American Confederation of Chemical Engineers



Roberto Andrade
Mexican Institute of Chemical Engineers



Maurice Allen
Institution of Engineers, Australia



Alanair I. Miller
Canadian Society for Chemical Engineers



Tim Dobbie
Society of Chemical Engineers, New Zealand



Gong Qi Yi
Chemical Industry & Engineering Society of China



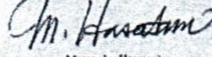
Jean-Claude Charpentier
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Masanobu Hassani
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